DAVID LEVY-BOOTH, PhD

AI for Biology · Canada · Remote & US-based roles
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SUMMARY

Full-stack AI engineer and computational scientist with the goal of bringing novel treatments to patients with rare diseases. I build biological design systems, turn large-scale experimental data into validated designs, and benchmark frontier models for biological safety and capability.

TECHNICAL SKILLS

AI/ML: Generative protein design models, protein language models, transformers, multi-GPU inference, distributed training & fine-tuning, LLMs, RAG & evaluation, model & benchmark evaluation (incl. biosafety), Weights & Biases, model versioning & promotion in production

Agentic AI: Claude Agents SDK, OpenAI Agents SDK, LangChain/LangGraph, MCP, vector databases (pgvector, Pinecone), knowledge graphs, agent & eval development

Languages & Frameworks: Python, R, TypeScript, SQL/NoSQL; PyTorch, FastAPI, React, Next.js

Infrastructure & MLOps: Kubernetes, Ray, GCP, AWS, NVIDIA DGX Cloud, Temporal, Modal, Terraform (IaC), Docker, CI/CD, Git; Nextflow, Snakemake, Argo; API development & auth

Domains: Protein engineering & structure, AAV capsid design, NGS (RNA-Seq, scRNA-Seq, WGS, ChIP-Seq), single-cell & spatial transcriptomics, metagenomics, genome assembly, variant calling, image segmentation & analysis

Leadership: Team & budget ownership, product leadership, roadmaps & KPIs, partnership negotiation, AI strategy & policy, cross-functional collaboration, mentorship

EXPERIENCE

Dyno Therapeutics · Watertown, MA (Remote) · 2022 – Present
Director, Engineering (2026–Present) · Associate Director, Computational Platform (2025–2026) · Principal Scientist (2023–2025) · Senior Scientist (2022–2023)

- Secured a **$5M+ NVIDIA compute partnership** as technical lead. Ported flow-matching training stack to NVIDIA DGX Cloud and directed the production migration, enabling release of Dyno Psi-1, a generative protein folding/binding model reaching **28% experimental success versus ~15% for leading open-source approaches**.
- Built Dyno's first computational product, Dyno Phi: weighted in-silico filters that lifted binder-design success to **42% versus ~11% for published filter sets**. Engineered the full stack myself: Python backend, Kubernetes/Terraform IaC, authenticated API (Clerk/JWT), and a Next.js/TypeScript front end. **100+ monthly users** across academia, industry, internal teams, and frontier AI labs; method publication in progress.
- Designed **millions of protein binders** (miniproteins, VHH antibodies, scFv) with generative models, and built the capsid-engineering models and platform behind Dyno's largest partnered programs, training on **55B sequence-function measurements across 344 in vivo and in vitro properties**.
- Led development of a **protein-design evaluation API** exposing viral-protein classification and design tasks, used by **OpenAI (GPT-5.5, GPT-5.6) and Anthropic (Claude Mythos, Fable 5)** to benchmark models against human experts. Ran **thousands of evaluations** under a paid frontier-lab partnership covering **biosafety**; results cited in published model cards.
- Launched Dyno's AI agent program, putting **4 agents into production** for scientific pipeline orchestration, biological design, and AI-assisted coding. Took the biological-design agent into **3 external partnerships** and direct work with **6 rare-disease foundation and parent groups**.
- Managed a **13-person team** across software, ML engineering, and data science with a **$2.5M annual compute budget**, and led the move to a modular Ray-based training and fine-tuning platform for rapid deployment onto new GPU hardware.
- Built NGS parsing service that turns a sequencing run into ready-to-train measurements from barcoded DNA and RNA library pools in **under an hour**, keeping pace with three NextSeq 2000 P4 flow cells and one NovaSeq each week.

University of British Columbia, Sequencing & Bioinformatics Core · Vancouver, BC · 2020 – 2022
Bioinformatics Specialist (Head of Bioinformatics)

- Launched automated bioinformatics workflows (NGS QC, RNA-Seq, WGS, ChIP-Seq, small RNA-Seq, scRNA-Seq, genome assembly, variant calling) and a service platform for UBC researchers and industry partners.
- **Doubled annual bioinformatics revenue** with custom genomics, 16S amplicon, CRISPR-screen, and single-cell and bulk transcriptomics solutions.

University of British Columbia, Microbiology & Immunology · Vancouver, BC · 2015 – 2020
Research Associate (2019–2020) · Hakai Institute Postdoctoral Fellow (2015–2019)

- Built a bioproduct discovery platform combining mass spectrometry, transcriptomics, protein characterization, and large-scale genome sequencing; secured **$500K+ in Genome BC and NSERC grants**.
- Published 7 peer-reviewed studies (4 first-author), including papers in PNAS and The ISME Journal.

Microbiome Insights Inc. · Vancouver, BC · 2018 – 2020
Consultant

- Developed metatranscriptomics and metagenomics pipelines behind a **$1.2M+ government research contract**, and built qPCR assays to monitor biotherapeutics through Phase I trials.

Natural Resources Canada, Pacific Forestry Centre · Victoria, BC · 2007 – 2009
Biologist, Government of Canada

EDUCATION

- **PhD, Microbial Ecology & Genomics**, University of British Columbia (2014)
- **MSc, Environmental Microbiology & Biotechnology**, University of Guelph (2007)
- **BA (Hons), Biology**, Wilfrid Laurier University (2005)

SELECTED PUBLICATIONS & TALKS

- Busia A et al. 2025. *Dyno Psi-1* white paper. Dyno Therapeutics (engineering). dynopsi.dynotx.com
- Invited talk: **Machine-guided design of AAVs for gene therapy delivery**, Data Infrastructure in Biotech Conference, Boston, 2024.
- Fetherolf MM*, Levy-Booth DJ* et al. 2020. Characterization of alkylguaiacol-degrading cytochromes P450 for the biocatalytic valorization of lignin. PNAS 117:25771–25778. (*co-first author)
- Levy-Booth DJ et al. 2022. Discovery of lignin-transforming bacteria and enzymes in thermophilic environments using stable isotope probing. The ISME Journal 16:1944–1956.

Full publication list (20+ papers): Google Scholar